

Week 5 Report: Exploratory Data Analysis (end) and Model Development (start)

Project: Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland

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What I did this week

This week I built the model pipeline. Each model is wrapped with scaling and, where used, SMOTE, so every step that learns from data only sees the training data. I then ran a first comparison of Logistic Regression, Decision Tree and Random Forest without any resampling.

Random Forest was already ahead with an F1 of 0.829, then Decision Tree at 0.786 and Logistic Regression at 0.683. All three have accuracy above 99.9%, which confirms that accuracy is not a useful measure on this data.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.9991	0.837	0.577	0.683
Decision Tree	0.9994	0.878	0.711	0.786
Random Forest	0.9995	0.954	0.732	0.829

Next week

Tune each model with cross-validation on the training data.